

BlastOpt Botswana - Optimization Report

AI-Driven Blasting Optimization Suite

1. Optimization Objectives & Constraints Summary

Objectives: Minimize D&B cost (\$/t) subject to vibration (PPV), flyrock, and fragmentation constraints.

- Max Ground Vibration Limit (PPV): 10.0 mm/s
- Max Flyrock Distance Limit: 120.0 m
- Target d50 Fragmentation Range: 120 - 320 mm

2. Top Recommended Blast Design Configurations

Rank	Burden (m)	Spacing (m)	Stemming (m)	PF (kg/m3)	d50 (mm)	PPV (mm/s)	Cost (\$/t)
1	7.41	3.99	6.97	0.605	319.7	10.00	\$0.92
2	6.79	3.00	5.60	0.520	330.7	6.57	\$1.06
3	7.86	3.00	5.60	0.520	338.9	7.39	\$0.97
4	8.93	3.00	2.00	0.840	255.5	12.01	\$1.14
5	8.93	3.00	5.60	0.520	346.1	8.18	\$0.90

3. Fragmentation Size Distribution Curve (Best Design)

Kuz-Ram Fragmentation Curve (Percent Passing vs Size)

